



Analytics Solutions Bootcamp: #02

Foundations of Data Management

Discover what data is and apply practical techniques to prepare reliable data for use.

Hello, everyone! Welcome to the second module of the Analytics Solutions Bootcamp. In this module, you will learn about the Foundations of Data Management. We will talk about the essential concepts in data management and learn how to work with data.

Topics to be covered



- Data management essential concepts
 - Data, data quality, data lifecycle, types of data
 - DAMA Wheel
- Sources of data: how is data created?
- Intro to data architecture: how is data stored & accessed?
- Tidy data principles: how is data prepped?
- Ethical principles of data: how do we respect data?
- Common data challenges

Data management essential concepts

Overview to data, data quality, data lifecycle, types of data, and the DAMA Wheel

Data management essential concepts

Concepts: Data and Information >>

Data and information is defined as follows.

- **Data** represents facts about the world. “raw material of information.”
- **Information** is “data in context”. Next stage of data in DIKW pyramid.

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Define words then provide the examples.

Example of data: age, gender, name, this bootcamp

Example of information: demographics of the bootcamp learners

Mention DIKW pyramid

[CLICK]

Data management essential concepts

Concepts: Data and Information >>

DIKW pyramid shows the relationship among data, information, knowledge, and wisdom.



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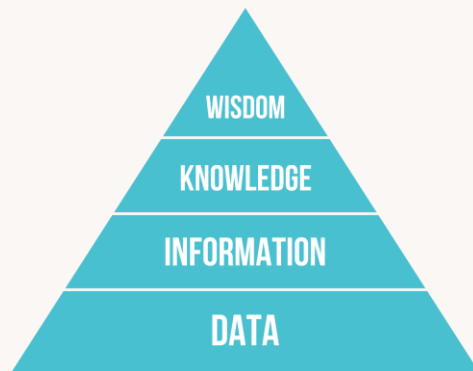
Concepts: Data and Information >>

Data: 36°C, 70% humidity

Information: It is hot and humid today.

Knowledge: Hot and humid weather increases the risk of dehydration.

Wisdom: I will bring a water bottle and wear light clothing.



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DIKW – Data, Information, Knowledge, Wisdom pyramid

Example 1: Weather

Data: 34°C, 70% humidity, 10 km/h wind

Information: *It is hot and humid today.*

Knowledge: *Hot and humid weather increases the risk of dehydration, so people should drink more water and avoid prolonged outdoor activities.*

Wisdom: *I will bring a water bottle, wear light clothing, and postpone my afternoon run until the evening.*

Example 2: Student Performance

Data: Quiz scores: 72, 68, 75, 70

Information: *The student's average quiz score is 71.25%, which is below the passing mark.*

Knowledge: *The student is consistently struggling with the subject and may need additional support or a different study strategy.*

Wisdom: *The teacher schedules remedial sessions and provides targeted practice exercises to improve the student's understanding.*

Example 3: Business Sales

Data: Daily sales: ₱52,000, ₱49,000, ₱75,000

Information: *Sales are significantly higher on weekends than on weekdays.*

Knowledge: *Weekend promotions and increased customer traffic contribute to higher sales.*

Wisdom: *The business increases staffing and launches promotional campaigns on weekends to maximize revenue.*

Concepts: Data and Information >>

Data is a form of information and information is a form of data.

Ex. A patient's blood pressure is: 150/95 mmHg

- For the nurse recording it, this is data.
- For the physician, together with the patient's history, it becomes information indicating elevated blood pressure.
- For a public health researcher, thousands of physicians' reports become data for studying hypertension trends.

Example 1: Temperature

Suppose a weather station records:

32.5

As **data**, 32.5 is simply a recorded value.

When you add context:

Temperature: **32.5°C**

Location: Tarlac

Time: 2:00 PM

It becomes **information** because the value now has meaning.

Now imagine a weather analyst receives:

Temperature in Tarlac at 2:00 PM: 32.5°C

For the analyst, this **information** becomes **data** that can be combined with thousands of other temperature readings to predict tomorrow's weather.

So:

The weather station produces **data**.

The report presents **information**.

The analyst treats that information as **data** for another analysis.

Example 2: Sales

A database contains:

Date

Product

Quantity

Jul 15Laptop8

Jul 16Laptop12

These are **data**.

A manager reads a dashboard saying:

Laptop sales increased by 50% compared to yesterday.

That is **information**.

Now suppose the regional director collects reports from 20 stores. Each store's summary ("sales increased by 50%") becomes another **data point** for regional analysis.

Example 3: Healthcare

A patient's blood pressure is:

150/95 mmHg

For the nurse recording it, this is **data**.

For the physician, together with the patient's history, it becomes **information** indicating elevated blood pressure.

For a public health researcher, thousands of physicians' reports become **data** for studying hypertension trends.

I don't want you guys to focus on the technical definitions and terms, but I want you to appreciate that data isn't just about figures and numbers. Data can be a form of information. An information can be a form of data.

Let's look at an excerpt from the book DMBOK.

[CLICK]



JOSH DEV PH

“Most people assume that, because data represents facts, *it is a form of truth* about the world and that the facts will fit together.

But ‘facts’ are not always simple or straightforward.”

DAMA-DMBOK: Data Management Body of Knowledge



Inquirer



PSA



World Bank Group



Inquirer

1st image: The news caused people, especially parents, to be worried about their routine deworming. The news article is framed poorly that it presented incorrect association.

2nd image: This new “achievement” of the Philippines garnered several reactions saying that they do not feel this growth.

3rd image: June 2026 inflation rates show 6.4%. It went down from 7.2 in March 2026 and 6.8 in April 2026. But the figure is significantly high compared to January 2026, with only 2% inflation and 1.4% inflation back in June 2025.

4th image: DTI says that a family of 5 can enjoy a Noche Buena worth 500 pesos.

Concepts: Data Quality



- As data professionals, we always crave for **high-quality data**.
- Poor data quality is costly: IBM estimated the cost of poor-quality data in the US in 2016 was **\$3.1 trillion**.
- Common **dimensions** of data quality.
 - Accuracy, completeness, consistency, integrity, reasonability, timeliness, uniqueness, and validity.

Data professionals always need high quality data.

- Data engineers will build their pipelines much easier
- Data analyst will be able to analyze their data much faster
- Data scientists will spend less time on cleansing and more time on statistical analysis and modeling

We spend more time on our work when we have poor quality of data. Why do we have losses due to poor data quality?

How do we measure the quality of our data? We have dimensions of data quality.

Concepts: Data Quality



Data Quality Dimension	Definition	Example
Accuracy	Data correctly represents the real-world object or event.	Customer birthdate stored as 1985-03-15 when the actual birthdate is 1995-03-15.
Completeness	All required data is present.	Customer record is missing email and phone number.
Consistency	Data is the same across systems and datasets.	Ex. 1: Customer status is Active in CRM but Inactive in billing system. Ex. 2: On customer gender, there are MALE, FEMALE, M, F, 0, 1
Integrity	Relationships between data elements are valid and maintained.	A sales transaction contains Customer ID 99999, which does not exist in the customer table.

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Data Quality Dimension

Definition

Good Example

Bad Example

Accuracy

- Data correctly represents the real-world object or event.
- Customer birthdate is **1995-03-15**, matching official records.
- Customer birthdate stored as **1985-03-15** when the actual birthdate is 1995-03-15.

Completeness

- All required data is present.
- Customer record contains name, email, phone number, and address.
- Customer record is missing email and phone number.

Consistency

- Data is the same across systems and datasets.
- Customer status is **Active** in both CRM and billing systems.

- Customer status is **Active** in CRM but **Inactive** in billing system.

Integrity

- Relationships between data elements are valid and maintained.
- Every sales transaction references a valid Customer ID.
- A sales transaction contains Customer ID **99999**, which does not exist in the customer table.

Concepts: Data Quality



Data Quality Dimension	Definition	Example
Reasonability	Data values make logical and business sense.	Employee age is 250 years old.
Timeliness	Data is available and up-to-date when needed.	Inventory was last updated three months ago.
Uniqueness	Each real-world entity is represented only once.	Three customer records exist for the same customer with slightly different spellings.
Validity	Data conforms to defined business rules and formats.	Email stored as user@email or 12345.

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Data Quality Dimension

Definition

Good Example 

Bad Example 

Reasonability

- Data values make logical and business sense.
- Employee age is **35 years old**.
- Employee age is **250 years old**.

Timeliness

- Data is available and up-to-date when needed.
- Inventory levels are updated every hour.
- Inventory was last updated three months ago.

Uniqueness

- Each real-world entity is represented only once.
- One customer record exists for Juan Dela Cruz.
- Three customer records exist for the same customer with slightly different

spellings.

Validity

- Data conforms to defined business rules and formats.
- Email stored as [user@email.com](#).
- Email stored as **user@email** or **12345**.

Data management essential concepts

Concepts: Data Quality



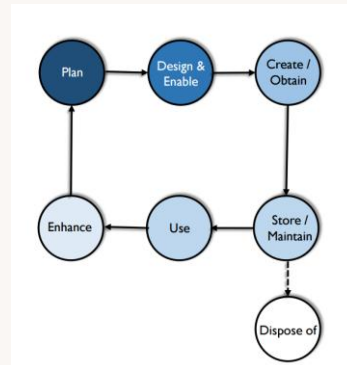
Data Profiling is a form of data analysis used to inspect data and assess quality. Data profiling uses statistical techniques to discover the true structure, content, and quality of a collection of data (Association et al. 2017b).

Data Profiling will help us inspect and assess the quality of our data.

Concepts: Data Lifecycle



The data lifecycle includes processes that create or obtain data, move, transform, and store it and enable it to be maintained and shared, and those that use or apply it, as well as those that dispose of it.



Concepts: Types of Data



There are various types of data. Some of which are as follows.

- By type: transactional, reference, master data, metadata
- By content: subject area, data domains
- By format: structured, semi-structured, unstructured

Examples:

- By type:
 - Transactional – daily sales data
 - Reference – table of store codes, product category
 - Master data – customer list, product catalog
 - Metadata – details about sales tables, last refresh date
- By content:
 - Subject area – human resources, retail
 - Data domains – employee domain, recruitment domain, customer domain, sales domain
- By format:
 - Structured – tabular, CSV, Excel files
 - Semi-structured – JSON, YAML, XML
 - Unstructured – PDF files, text files, images, audio/video files

Sources of data: how is data created?

Files and unstructured data, API, application databases, messages and streams

Sources of data: how is data created?

Data creation



Data is created by in two ways: **analog** and **digital**.

- Analog data creation occurs in the real world (vocal speech, sign language, writing on paper, playing an instrument).
- Digital data is created by converting analog data to digital form (mobile texting app, credit card transaction).

Sources of data: how is data created?

Source systems



Here are the main sources systems where data is created or sourced from.

1. Files and unstructured data
2. APIs
3. Application databases
4. Message and streams

Data architecture: how data is stored & accessed?

A brief overview on data warehouse, data lake, and data lakehouse

Data architecture: how data is stored & accessed?

Definition



Data architecture is the design of systems to support the evolving data needs of an enterprise, achieved by flexible and reversible decisions reached through a careful evaluation of trade-offs.

Data architecture: how data is stored & accessed?

Data architecture types >>

Here are main types of data architecture designs that are used by organizations.

1. Data warehouse is a central data hub for reporting and analysis.
2. Data lake stores the raw and unprocessed form of data.
3. Data lakehouse is the hybrid of data warehouse and data lake.

Tidy data principles: how data is prepped?

General practice on preparing your data for analysis

Tidy data principles: how data is prepped?

Definition



Tidy data is a standard way of mapping the meaning of a dataset to its structure. A dataset is messy or tidy depending on how rows, columns and tables are matched up with observations, variables and types. In tidy data:

1. Each variable forms a column.
2. Each observation forms a row.
3. Each type of observational unit forms a table.

Tidy data principles: how data is prepped?

Tidying messy datasets



1. Column headers are values, not variable names.
2. Multiple variables are stored in one column.
3. Variables are stored in both rows and columns.
4. Multiple types of observational units are stored in the same table.
5. A single observational unit is stored in multiple tables.

Tidy data principles: how data is prepped?

Issue 1 example:



Column headers are values, not variable names. We need to “flatten” or “unpivot” the data to make it much easier to analyze.

religion	<\$10k	\$10-20k	\$20-30k	\$30-40k	\$40-50k	\$50-75k
Agnostic	27	34	60	81	76	137
Atheist	12	27	37	52	35	70
Buddhist	27	21	30	34	33	58
Catholic	418	617	732	670	638	1116
Don't know/refused	15	14	15	11	10	35
Evangelical Prot	575	869	1064	982	881	1486
Hindu	1	9	7	9	11	34
Historically Black Prot	228	244	236	238	197	223
Jehovah's Witness	20	27	24	24	21	30
Jewish	19	19	25	25	30	95

Table 4: The first ten rows of data on income and religion from the Pew Forum. Three columns, \$75-100k, \$100-150k and >150k, have been omitted

religion	income	freq
Agnostic	<\$10k	27
Agnostic	\$10-20k	34
Agnostic	\$20-30k	60
Agnostic	\$30-40k	81
Agnostic	\$40-50k	76
Agnostic	\$50-75k	137
Agnostic	\$75-100k	122
Agnostic	\$100-150k	109
Agnostic	>150k	84
Agnostic	Don't know/refused	96

Table 6: The first ten rows of the tidied Pew survey dataset on income and religion. The column has been renamed to *income*, and *value* to *freq*.

Tidy data principles: how data is prepped?

Issue 2 example:



Multiple variables are stored in one column. Example, count of male aged 0-14 are displayed as one column. This should be flattened.

country	year	m014	m1524	m2534	m3544	m4554	m5564	m65	mu	f014
AD	2000	0	0	1	0	0	0	0	—	—
AE	2000	2	4	4	6	5	12	10	—	3
AF	2000	52	228	183	149	129	94	80	—	93
AG	2000	0	0	0	0	0	0	1	—	1
AL	2000	2	19	21	14	24	19	16	—	3
AM	2000	2	152	130	131	63	26	21	—	1
AN	2000	0	0	1	2	0	0	0	—	0
AO	2000	186	999	1003	912	482	312	194	—	247
AR	2000	97	278	594	402	419	368	330	—	121
AS	2000	—	—	—	—	1	1	—	—	—

Table 9: Original TB dataset. Corresponding to each 'm' column for males, there is also an 'f' column for females, f1524, f2534 and so on. These are not shown to conserve space. Note the mixture of 0s and missing values (—). This is due to the data collection process and the distinction is important for this dataset.

country	year	column	cases	country	year	sex	age	cases
AD	2000	m014	0	AD	2000	m	0-14	0
AD	2000	m1524	0	AD	2000	m	15-24	0
AD	2000	m2534	1	AD	2000	m	25-34	1
AD	2000	m3544	0	AD	2000	m	35-44	0
AD	2000	m4554	0	AD	2000	m	45-54	0
AD	2000	m5564	0	AD	2000	m	55-64	0
AD	2000	m65	0	AD	2000	m	65+	0
AE	2000	m014	2	AE	2000	m	0-14	2
AE	2000	m1524	4	AE	2000	m	15-24	4
AE	2000	m2534	4	AE	2000	m	25-34	4
AE	2000	m3544	6	AE	2000	m	35-44	6
AE	2000	m4554	5	AE	2000	m	45-54	5
AE	2000	m5564	12	AE	2000	m	55-64	12
AE	2000	m65	10	AE	2000	m	65+	10
AE	2000	f014	3	AE	2000	f	0-14	3

(a) Molten data

(b) Tidy data

Table 10: Tidying the TB dataset requires first melting, and then splitting the column column into two variables: sex and age.

Tidy data principles: how data is prepped?

Issue 3 example:



Variables are stored in both rows and columns. Sex and age variables need to split.

country	year	m014	m1524	m2534	m3544	m4554	m5564	m65	mu	f014
AD	2000	0	0	1	0	0	0	0	—	—
AE	2000	2	4	4	6	5	12	10	—	3
AF	2000	52	228	183	149	129	94	80	—	93
AG	2000	0	0	0	0	0	0	1	—	1
AL	2000	2	19	21	14	24	19	16	—	3
AM	2000	2	152	130	131	63	26	21	—	1
AN	2000	0	0	1	2	0	0	0	—	0
AO	2000	186	999	1003	912	482	312	194	—	247
AR	2000	97	278	594	402	419	368	330	—	121
AS	2000	—	—	—	—	1	1	—	—	—

Table 9: Original TB dataset. Corresponding to each 'm' column for males, there is also an 'f' column for females. f1524, f2534 and so on. These are not shown to conserve space. Note the mixture of 0s and missing values (—). This is due to the data collection process and the distinction is important for this dataset.

country	year	column	cases	country	year	sex	age	cases
AD	2000	m014	0	AD	2000	m	0-14	0
AD	2000	m1524	0	AD	2000	m	15-24	0
AD	2000	m2534	1	AD	2000	m	25-34	1
AD	2000	m3544	0	AD	2000	m	35-44	0
AD	2000	m4554	0	AD	2000	m	45-54	0
AD	2000	m5564	0	AD	2000	m	55-64	0
AD	2000	m65	0	AD	2000	m	65+	0
AE	2000	m014	2	AE	2000	m	0-14	2
AE	2000	m1524	4	AE	2000	m	15-24	4
AE	2000	m2534	4	AE	2000	m	25-34	4
AE	2000	m3544	6	AE	2000	m	35-44	6
AE	2000	m4554	5	AE	2000	m	45-54	5
AE	2000	m5564	12	AE	2000	m	55-64	12
AE	2000	m65	10	AE	2000	m	65+	10
AE	2000	f014	3	AE	2000	f	0-14	3

(a) Molten data

(b) Tidy data

Table 10: Tidying the TB dataset requires first melting, and then splitting the column column into two variables: `sex` and `age`.

Tidy data principles: how data is prepped?

Issue 4 example:



Multiple types in one table. We need to flatten the data and then normalize.

year	artist	track	time	date.entered	wk1	wk2	wk3
2000	2 Pac	Baby Don't Cry	4:22	2000-02-26	87	82	72
2000	2Ge+her	The Hardest Part Of ...	3:15	2000-09-02	91	87	92
2000	3 Doors Down	Kryptonite	3:53	2000-04-08	81	70	68
2000	98°0	Give Me Just One Nig...	3:24	2000-08-19	51	39	34
2000	A*Teens	Dancing Queen	3:44	2000-07-08	97	97	96
2000	Aaliyah	I Don't Wanna	4:15	2000-01-29	84	62	51
2000	Aaliyah	Try Again	4:03	2000-03-18	59	53	38
2000	Adams, Yolanda	Open My Heart	5:30	2000-08-26	76	76	74

Table 7: The first eight Billboard top hits for 2000. Other columns not shown are `wk4`, `wk5`, ..., `wk75`.

year	artist	time	track	date	week	rank
2000	2 Pac	4:22	Baby Don't Cry	2000-02-26	1	87
2000	2 Pac	4:22	Baby Don't Cry	2000-03-04	2	82
2000	2 Pac	4:22	Baby Don't Cry	2000-03-11	3	72
2000	2 Pac	4:22	Baby Don't Cry	2000-03-18	4	77
2000	2 Pac	4:22	Baby Don't Cry	2000-03-25	5	87
2000	2 Pac	4:22	Baby Don't Cry	2000-04-01	6	94
2000	2 Pac	4:22	Baby Don't Cry	2000-04-08	7	99
2000	2Ge+her	3:15	The Hardest Part Of ...	2000-09-02	1	91
2000	2Ge+her	3:15	The Hardest Part Of ...	2000-09-09	2	87
2000	2Ge+her	3:15	The Hardest Part Of ...	2000-09-16	3	92
2000	3 Doors Down	3:53	Kryptonite	2000-04-08	1	81
2000	3 Doors Down	3:53	Kryptonite	2000-04-15	2	70
2000	3 Doors Down	3:53	Kryptonite	2000-04-22	3	68
2000	3 Doors Down	3:53	Kryptonite	2000-04-29	4	67
2000	3 Doors Down	3:53	Kryptonite	2000-05-06	5	66

Table 8: First fifteen rows of the tidied billboard dataset. The `date` column does not appear in the original table, but can be computed from `date.entered` and `week`.

Tidy data principles: how data is prepped?

Issue 4 example:



Multiple types in one table. We need to flatten the data and then normalize.

id	artist	track	time	id	date	rank
1	2 Pac	Baby Don't Cry	4:22	1	2000-02-26	87
2	2Ge+her	The Hardest Part Of ...	3:15	1	2000-03-04	82
3	3 Doors Down	Kryptonite	3:53	1	2000-03-11	72
4	3 Doors Down	Loser	4:24	1	2000-03-18	77
5	504 Boyz	Wobble Wobble	3:35	1	2000-03-25	87
6	98~0	Give Me Just One Nig...	3:24	1	2000-04-01	94
7	A*Teens	Dancing Queen	3:44	1	2000-04-08	99
8	Aaliyah	I Don't Wanna	4:15	2	2000-09-02	91
9	Aaliyah	Try Again	4:03	2	2000-09-09	87
10	Adams, Yolanda	Open My Heart	5:30	2	2000-09-16	92
11	Adkins, Trace	More	3:05	3	2000-04-08	81
12	Aguilera, Christina	Come On Over Baby	3:38	3	2000-04-15	70
13	Aguilera, Christina	I Turn To You	4:00	3	2000-04-22	68
14	Aguilera, Christina	What A Girl Wants	3:18	3	2000-04-29	67
15	Alice DeeJay	Better Off Alone	6:50	3	2000-05-06	66

Table 13: Normalised billboard dataset split up into song dataset (left) and rank dataset (right). First 15 rows of each dataset shown; **genre** omitted from song dataset, **week** omitted from rank dataset.

Tidy data principles: how data is prepped?

Issue 5 example:



One type in multiple tables

It's also common to find data values about a single type of observational unit spread out over multiple tables or files. These tables and files are often split up by another variable, so that each represents a single year, person, or location. As long as the format for individual records is consistent, this is an easy problem to fix.

Ethical principles of data: how do we respect data?

Introduction to data privacy and data ethics

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Ethical principles of data: how do we respect data?

Ethical Principles for Data



DMBOK presents that Belmont Principles for medical research may be adapted in Information Management disciplines.

1. Respect for persons
2. Beneficence
3. Justice

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Data Engineers can have a black and white guide.

Data Analysts and Data Scientists could play a big role in this area. Example, campaigns launched to specific group of people (ie. online gambling apps).

Ethical principles of data: how do we respect data?

Processing of Personal Info



Chapter 3 of the RA 10173 Data Privacy Act of 2012 includes the following provisions.

1. The General Data Privacy Principles emphasizes **transparency**, **legitimate purpose**, and **proportionality**.
2. Adhere to the **criteria for lawful processing** of personal information.
3. Processing of sensitive personal information **shall be prohibited** except for specific cases.

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Transparency

Individuals should know what data is collected and why. [\[privacy.gov.ph\]](https://privacy.gov.ph)

Legitimate Purpose

Data should be collected and processed for a lawful, specific purpose.

[\[privacy.gov.ph\]](https://privacy.gov.ph)

Proportionality

Only collect and process data necessary for the intended purpose.

[\[privacy.gov.ph\]](https://privacy.gov.ph)

Classroom Example

-  Collecting a customer's religion to process a food delivery order
-  Collecting a delivery address to fulfill the order

Ethical principles of data: how do we respect data?

Rights of the Data Subject



Chapter 4 of the RA 10173 Data Privacy Act of 2012 **enumerates the rights of the data subject**. Data subject refers to an individual whose personal information is processed.

Ethical principles of data: how do we respect data?

Responsible data handling



Practical habits every data professional should practice.

1. Collect only necessary data
2. Restrict access to sensitive data
3. Avoid sharing files unnecessarily
4. Store data securely
5. Dispose of data properly

Ethical principles of data: how do we respect data?

Ethics in Analytics



Move beyond legal compliance.

1. Is the analysis fair?
2. Could it harm individuals?
3. Are the conclusions misleading?
4. Would I be comfortable explaining this analysis to the affected people?

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